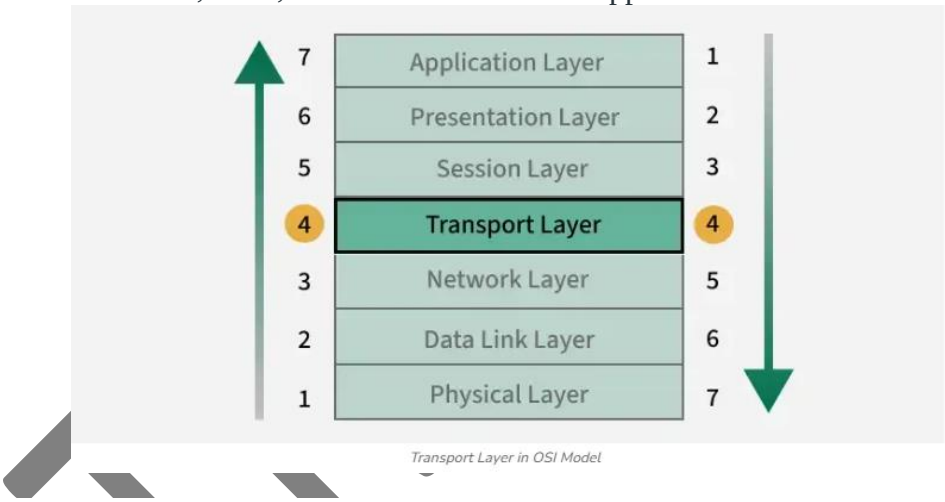
Unit II
Data Communication & Network

Transport Layer in OSI Model

The Transport Layer is the OSI Layer 4 responsible for providing reliable, efficient, and ordered end-to-end communication between applications on different hosts.

- Operates at Layer 4 (Transport Layer) of the OSI model
- Enables process-to-process communication using port numbers
- Ensures reliability through error control, sequencing, and retransmission
- Supports flow control to prevent receiver overload
- Uses protocols like TCP, UDP, and SCTP for different application needs



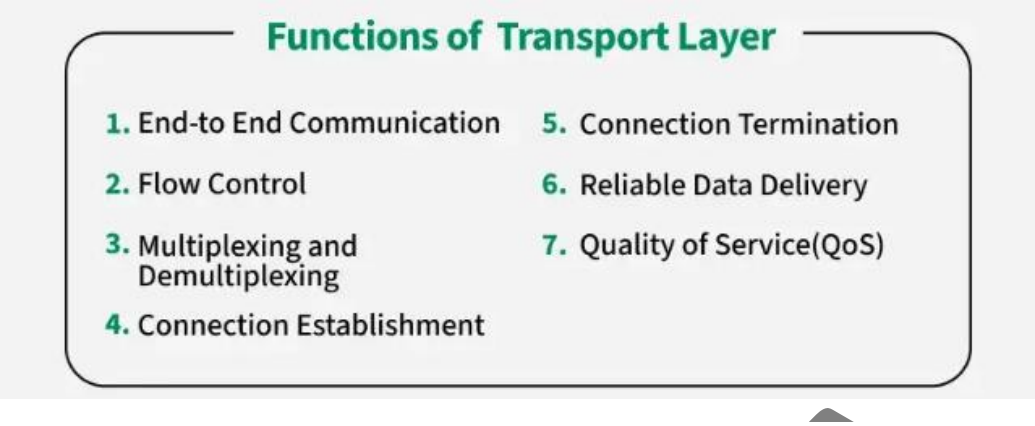
Transport Layer in OSI Model

Functions of Transport Layer

The Transport Layer is responsible for end-to-end communication of data packets. It provides a number of important functions that are responsible for reliable, efficient, and organized data transfer between host systems in a networked environment.

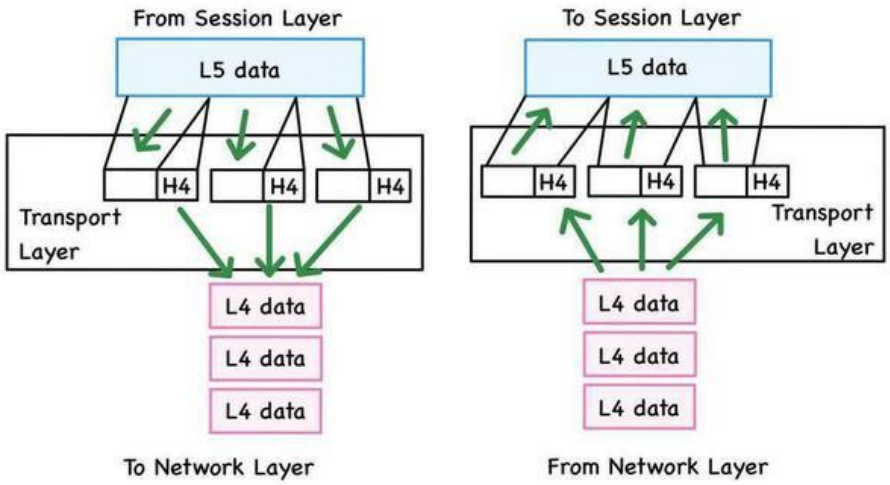
The primary functions of the Transport Layer are:

- Provides end-to-end data delivery
- Ensures reliable and ordered data transfer
- Performs error detection and recovery
- Controls data flow to prevent receiver overload
- Supports multiplexing and demultiplexing using port numbers



Working of Transport Layer

The Transport Layer provides logical communication between processes on different hosts meaning that even though data travels across various physical networks, the communicating applications perceive a direct, reliable link.



- Implemented only in end systems, not in intermediate routers.
- Uses port numbers to identify sending and receiving applications.
- Supports process-to-process delivery, enabling multiple applications to share a single network connection.
- Performs multiplexing and demultiplexing using port numbers to direct data to the correct process.
- Divides data from upper layers into segments (TCP) or datagrams (UDP) and adds necessary headers.
- Handles error detection, retransmission, and sequencing to maintain reliable communication.
- Coordinates flow control to ensure the receiver is not overloaded.
- Communicates with the Network Layer, which handles addressing and routing, to send data across networks.
- At the receiving end, it removes headers, reassembles data, and passes it to the appropriate application.

3-Way Handshake

The 3-Way Handshake ensures both client and server are ready before data transmission begins:

Step 1: SYN (Client -> Server)

- Client sends a TCP segment with SYN=1, including its ISN (Initial Sequence Number).
- Marks the request to initiate a connection

Step 2: SYN-ACK (Server -> Client)

- Server replies with SYN=1 & ACK=1, containing its own ISN and ACK = client_ISN + 1.

- Confirms receipt of client's SYN and initiates its own sync

Step 3: ACK (Client -> Server)

- Client sends an ACK=1 segment with ACK = server_ISN + 1.
- Completes synchronization; connection enters ESTABLISHED state

Why not just two steps?

With only SYN and SYN-ACK, neither the client nor the server can be sure that the other has successfully received the messages. The third ACK provides mutual confirmation, ensuring both sides are ready before data transfer begins.

Transport Layer Protocols

Transport Layer Protocol uses different protocol for the better communication between two ends uses of protocol may differ from specifications. Below mention are some protocols used in Transport Layer

1. Transmission Control Protocol(TCP)

- **TCP** is connection-oriented Protocol.
- TCP is reliable protocol.
- As TCP is connection-oriented protocol, so first the connection is established between two ends and then data is transferred and then the connection is terminated after all data being sent.

2. User Datagram Protocol (UDP)

- UDP is not reliable protocol
- The protocol UDP is connectionless.
- When speed and size are more important than security and dependability, this kind of protocol is employed.
- The data from the higher layer is supplemented with transport-level addresses, checksum error control, and length information by UDP, an end-to-end transport level protocol.
- A user datagram is the packet that the UDP protocol generates.

3. Stream Control Transmission Protocol (SCTP)

- Many Internet applications use SCTP to perform transport layer duties, similar to User Datagram Protocol (UDP) and Transmission Control Protocol (TCP).
- On top of a connectionless packet network like IP, SCTP is a dependable transport protocol that facilitates data transfer over the network in scenarios involving one or more IP addresses.

Difference Between TCP and UDP at Transport Layer

Choosing between TCP and UDP depends on the application's need for reliability versus speed and efficiency.

TCP vs UDP

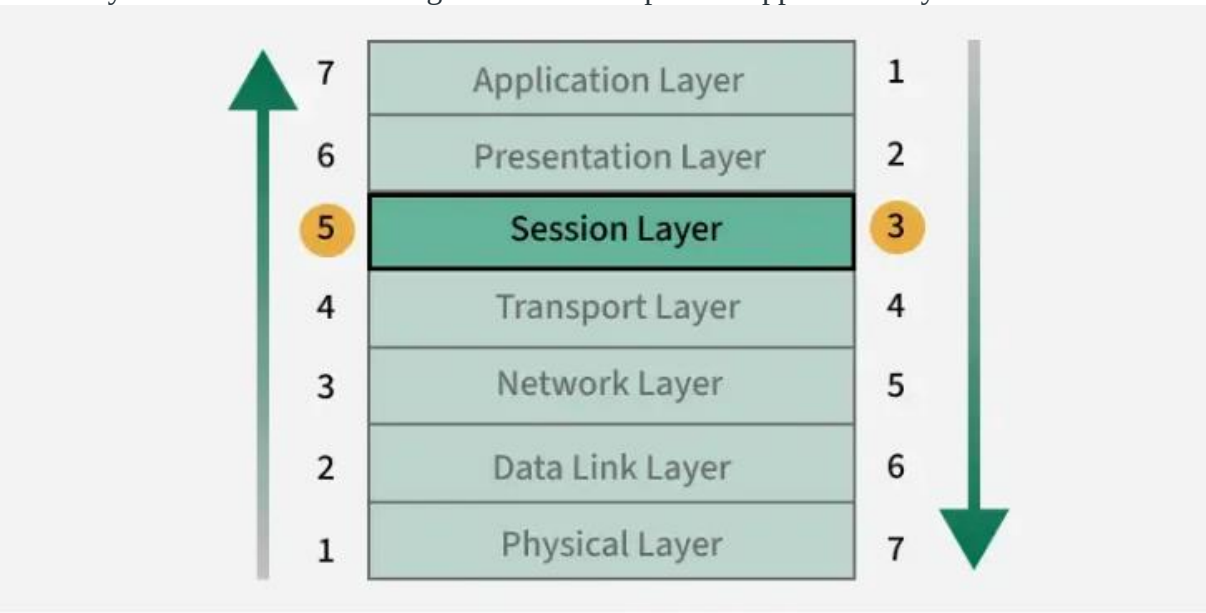
TCP	UDP
TCP is a connection-oriented protocol	UDP is the connection-less protocol

TCP	UDP
TCP supports error-checking mechanisms.	UDP has only the basic error-checking mechanism using checksums.
An acknowledgment segment is present.	No acknowledgment segment.
TCP is slower than UDP	UDP is faster, simpler, and more efficient than TCP.
Retransmission of lost packets is possible in TCP, but not in UDP.	There is no retransmission of lost packets in the User Datagram Protocol (UDP)
TCP has a (20-60) bytes variable length header.	The header length is fixed of 8 bytes.

Session Layer in OSI model

The Session Layer is the 5th layer of the OSI model that establishes, manages, synchronizes, and terminates communication sessions between applications on different devices.

- Operates at Layer 5 (Session Layer) of the OSI model
- Manages session setup, maintenance, and termination
- Controls dialogue (who sends/receives and when)
- Provides synchronization and recovery mechanisms
- Many of its functions are integrated into Transport or Application layers in modern TCP/IP networks



Session Layer

Role of the Session Layer
The Session Layer ensures that two communicating devices can establish a meaningful dialogue, exchange data in an organized manner and properly close the session when communication is complete.

- It provides mechanisms for session setup, management and termination.
- It ensures that communication remains synchronized and reliable, even during long or complex data transfers.
- It handles dialogue control, deciding whose turn it is to send or receive data.

Note: *In modern TCP/IP networks, some of its functions (like session release and dialogue control) are handled at the Transport Layer (TCP) or Application Layer, reducing the Session Layer's independent role.*

Key Functions of the Session Layer
- **Session Establishment:** Initiates and negotiates communication parameters (e.g., authentication, duplex mode).
- **Communication Synchronization:** Keeps data streams in order using checkpoints.
- **Activity & Dialog Management:** Controls turns, prevents collisions, and avoids duplication.
- **Resynchronization & Recovery:** Recovers from failures using synchronization points.
- **Session Termination:** Gracefully ends communication after all data is exchanged.

Working of the Session Layer
Working of Session Layer

- Establishes and negotiates session parameters (e.g., authentication, duplex mode).
- Manages token-based dialogue control to avoid collisions.
- Inserts synchronization checkpoints for recovery from failures.
- Ensures data integrity by reducing duplication or message loss.
- Gracefully terminates sessions after confirming all data has been exchanged.

Session Layer Protocols
Several protocols and technologies operate at the Session Layer:

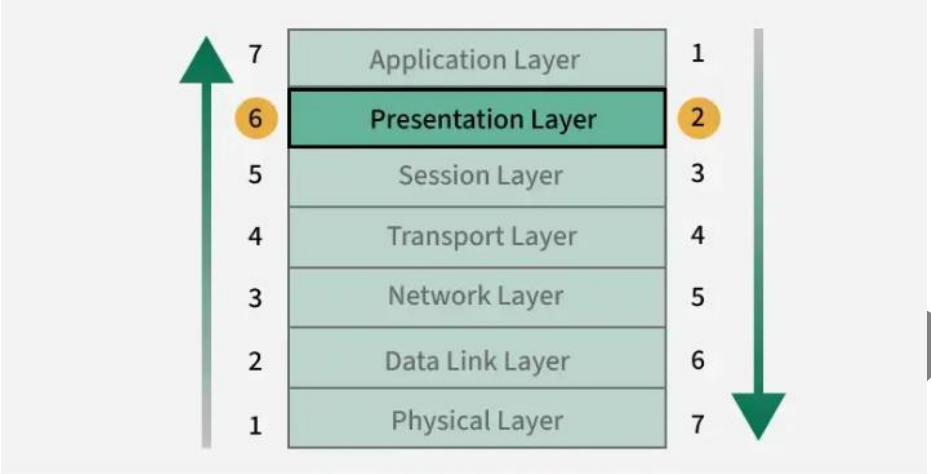
- **AppleTalk Data Stream Protocol (ADSP):** Developed by Apple for LAN communication with self-configuration support.
- **Real-time Transport Control Protocol (RTCP):** Provides QoS feedback for RTP-based multimedia sessions.
- **Point-to-Point Tunnelling Protocol (PPTP):** Enables Virtual Private Networks (VPNs) over TCP/IP.
- **Password Authentication Protocol (PAP):** Provides password-based user authentication in PPP connections.
- **Remote Procedure Call Protocol (RPCP):** Allows a program to execute procedures in another address space (client-server interaction).
- **Sockets Direct Protocol (SDP):** Supports socket communication over RDMA-enabled networks.

Devices Associated with the Session Layer
- **Firewalls:** Monitor and control sessions for security.
- **Proxy Servers:** Act as intermediaries, managing sessions between clients and servers.
- **Session Border Controllers (SBCs):** Secure and manage VoIP sessions.
- **Application Servers:** Create and maintain user sessions for applications.

Presentation Layer in OSI model

The Presentation Layer is the 6th layer of the OSI model that translates, encrypts, compresses, and formats data so that communicating systems can correctly understand each other.

- Operates at Layer 6 (Presentation Layer) of the OSI model
- Also called the Translation Layer or Syntax Layer
- Ensures proper data format, syntax, and semantics
- Provides encryption/decryption for security
- Applies compression to optimize bandwidth



Presentation Layer in OSI Model

Role of the Presentation Layer

When the Application Layer generates data, the Presentation Layer converts it into a standard form that can be transmitted across the network. Similarly, when data is received, it translates it into a format the receiving system can process.

- Maintains proper syntax and semantics of the data.
- Provides encryption and decryption for secure communication.
- Applies compression techniques to optimize bandwidth usage.
- Ensures compatibility between different systems and devices.

Functions of the Presentation Layer

Below are the functions of presentation layer:

- **Data Translation**: Converts data from the application's format into a standard network format (e.g., character encoding like ASCII ↔ Unicode) so different systems can understand each other.
- **Data Compression**: Reduces the size of data before transmission using compression techniques, which saves bandwidth and improves transmission speed.
- **Data Encryption/Decryption**: Encrypts data at the sender's side to protect it from unauthorized access and decrypts it at the receiver's side to restore readable information.
- **Syntax and Semantics Management**: Ensures that data structure (syntax) and meaning (semantics) remain consistent between communicating systems, preventing misinterpretation.
- **Transfer Syntax Negotiation**: Agrees on common data representation rules (format, encoding, compression method) before communication begins to ensure compatibility.
- **Interoperability**: Enables communication between heterogeneous systems by handling differences in operating systems, data formats, and architectures.

Services Provided by the Presentation Layer

The Presentation Layer ensures smooth and secure data exchange by providing the following services:

- **Compression**: Reduces data size for faster transmission.
- **Encryption/Decryption**: Protects data from unauthorized access.
- **Format Translation**: Converts application-specific data into a standard format.
- **Compatibility**: Makes communication possible between different operating systems and platforms.

Working of the Presentation Layer

- The Presentation Layer works as an intermediary between the Application Layer (Layer 7) and the Session Layer (Layer 5).
- At the sender's end, it formats, encrypts and compresses data received from the Application Layer before sending it to the Session Layer.
- At the receiver's end, it decrypts, decompresses and translates the data into a readable form before delivering it to the Application Layer.

Presentation Layer Protocols

- **Apple Filing Protocol (AFP):** File services protocol for macOS.
- **Lightweight Presentation Protocol (LPP):** Provides ISO presentation services over TCP/IP stacks.
- **NetWare Core Protocol (NCP):** Used in Novell NetWare for file and print services.
- **Network Data Representation (NDR):** Defines data types and representations for network communication.
- **External Data Representation (XDR):** Standard for describing and encoding data across different architectures.
- **Secure Socket Layer (SSL):** Provides encryption and secure communication between web browsers and servers.
- **Transport Layer Security (TLS):** The modern, more secure successor to SSL.

Presentation Layer Attacks

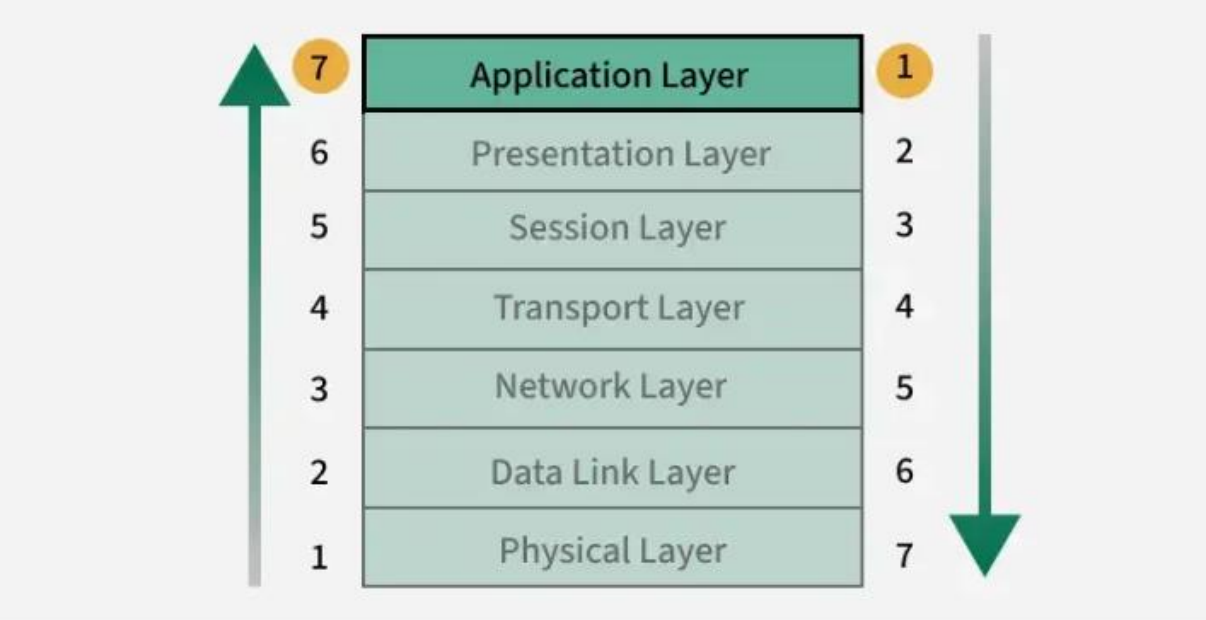
Since this layer deals with data formatting, compression and encryption, it is often targeted by attackers. Common attacks include:

- **Man-in-the-Middle (MITM) Attacks:** Interception of communication to steal sensitive data.
- **SSL/TLS Downgrade Attacks:** Forcing weaker encryption protocols.
- **Certificate Spoofing:** Using fake certificates to impersonate trusted entities.
- **Code Injection:** Exploiting vulnerabilities in data parsing or formatting.

Application Layer in OSI Model

The Application Layer is the topmost layer (Layer 7) of the OSI Model that enables direct interaction between end-user applications and the network.

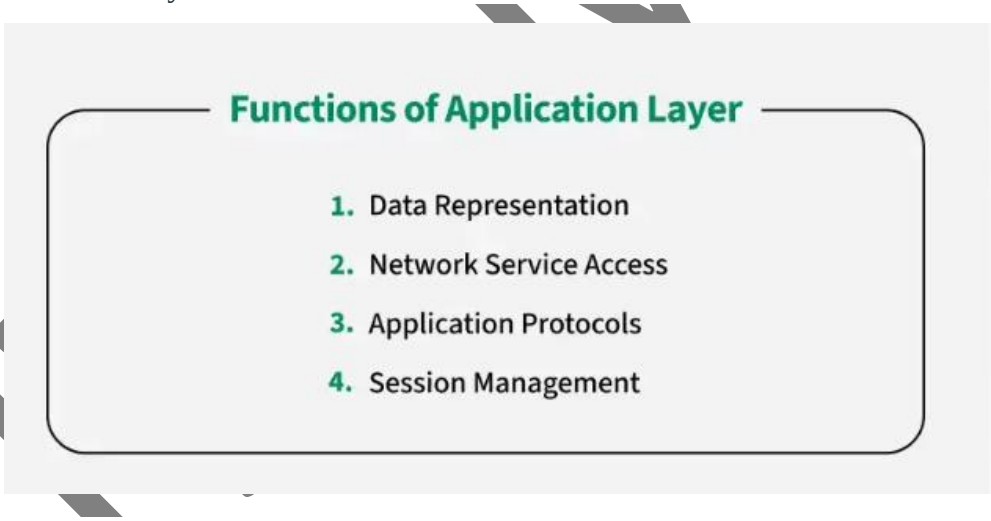
- Closest layer to the end user
- Provides network services to applications
- Enables file transfer, email, web access, and remote login
- Acts as an interface between the user software and the network



Application Layer in OSI Model

Functions

As the highest layer, the Application Layer performs several functions to ensure smooth communication between applications and systems:



1. Data Representation

- Converts user-entered information into a standard format before transmission and interprets received data back into a user-friendly form.
- **Example:** Converting text, images or files into network-compatible formats like ASCII, JPEG or HTML.

2. Network Service Access

- Provides access to various network services directly to the user.
- Allows applications like email, file transfer and remote login to interact with the underlying network.
- **Example:** Using email clients (Outlook, Gmail) to access mail services or using FTP to download/upload files.

3. Application Protocols

- Defines a set of rules and procedures that applications use to communicate over a network.
- These protocols handle message formatting, data exchange and service requests.

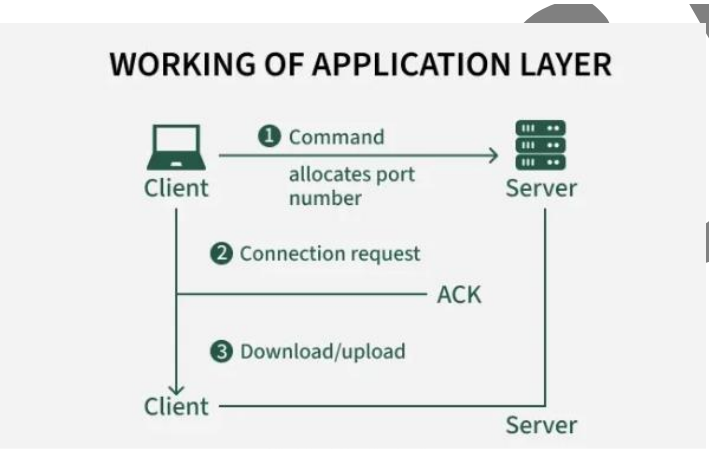
- **Example:** HTTP for web browsing, FTP for file transfer, SMTP for emails, DNS for domain name resolution.

4. Session Management

- Establishes, manages and terminates communication sessions between applications.
- Maintains synchronization during data exchange and ensures a proper connection is closed after the communication ends.
- **Example**: Logging into a remote server using Telnet or SSH, where a session is created and terminated after logout.

Working

Once the data exchange is complete, the client or server can terminate the session and the connection is closed gracefully.



Working of Application Layer

- At first, the client sends a command to the server and when server receives that command, it allocates a port number to client.
- Thereafter, the client sends an initiation connection request to server and when server receives request, it gives acknowledgement (ACK) to client through client has successfully established a connection with the server.
- Therefore, now client has access to server through which it may either ask server to send any types of files or other documents or it may upload some files or documents on server itself.

Key Concepts of the Application Layer

- Client-Server Model: Enables clients to request services or resources from a server over a network.
- World Wide Web (WWW): Provides access to interconnected web pages and resources over the internet.
- E-Mail: Enables users to send and receive electronic messages over a network.
- Content Distribution Network(CDN): Distributes web content through geographically distributed servers to improve delivery speed and reduce latency.

Services Provided

- Services Provided: Lists the key services offered by the system, application, or protocol.
- Define the communication rules followed by participating entities.
- Specify the format and type of messages exchanged between source and destination.
- Establish the structure and syntax of application-level data.

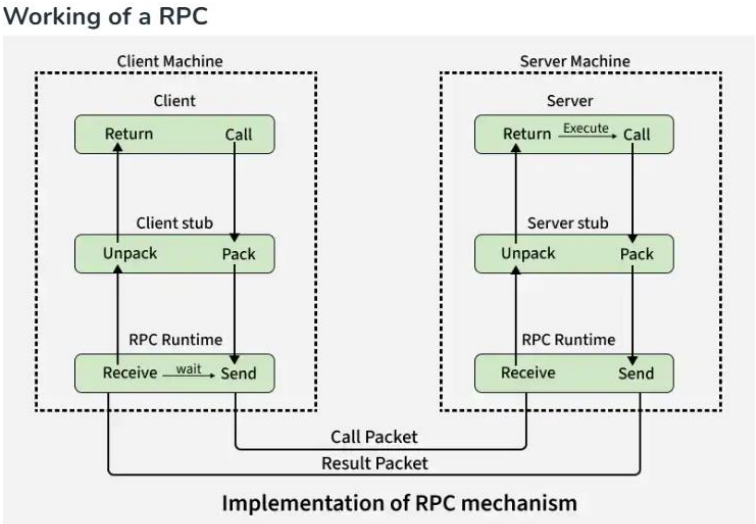
- Determine how requests are initiated and how responses are handled.
- Manage interaction between the application layer and lower network layers.

Protocols

- HyperText Transfer Protocol(HTTP): Used for web communication (Port 80).
- Domain Name System (DNS): Translates domain names into IP addresses (Port 53).
- TELNET: Provides remote login services (Port 23).
- Dynamic Host Configuration Protocol (DHCP): Assigns IP addresses to hosts dynamically (Ports 67 & 68).
- File Transfer Protocol (FTP): Facilitates file transfer between systems (Ports 20 for data, 21 for control).
- Simple Mail Transfer Protocol (SMTP): Handles sending of emails (Ports 25 & 587).
- Network File System (NFS): Enables remote access to files as if they were local (Port 2049).
- Simple Network Management Protocol (SNMP): Manages and monitors devices in a network (Ports 161 & 162).
- HyperText Transfer Protocol Secure(HTTPS): Provides secure web communication (Port 443).
- Post Office Protocol v3 (POP3): Retrieves emails from a mail server (Port 110).
- Internet Message Access Protocol(IMAP): Retrieves and manages emails on a mail server (Port 143).
- Lightweight Directory Access Protocol(LDAP): Provides access to and manages directory services (Port 389).
- Network Time Protocol(NTP): Synchronizes clocks across networked systems (Port 123).
- Trivial File Transfer Protocol(TFTP): Provides simple file transfer without authentication (Port 69).
- Network News Transfer Protocol(NNTP): Transfers and distributes Usenet news articles (Port 119).
- Message Queuing Telemetry Transport(MQTT): Enables lightweight messaging for IoT devices (Port 1883).
- Session Initiation Protocol(SIP): Establishes and manages communication sessions (Port 5060).
- Server Message Block(SMB): Provides file and printer sharing over a network (Port 445).

Remote Procedure Call (RPC)

Remote Procedure Call (RPC) is a way for a program to run a function on another computer in a network as if it were local. The client sends the request (with arguments) to the server, the server executes the function, and the result is sent back. RPC hides the details of networking, so developers can think in terms of normal function calls instead of complex network operations.

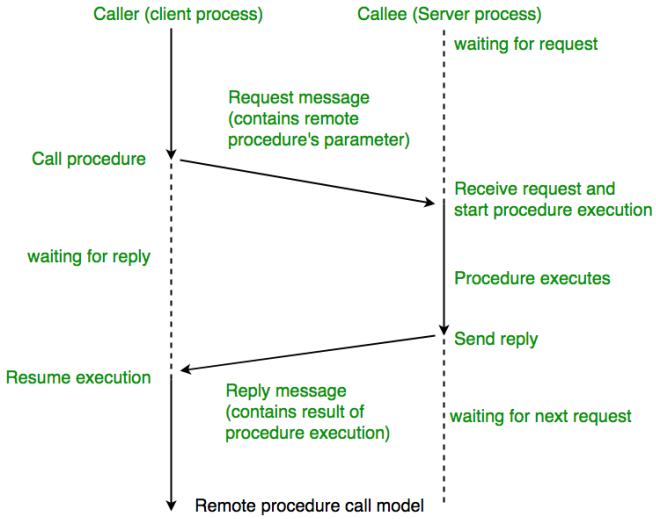


Remote Procedure Call (RPC)

How RPC Works (Step by Step)

1. **Client Calls Stub:** The client calls a local procedure (stub) as if it were normal.
2. **Marshalling**: The stub packs (marshals) all input parameters into a message.
3. **Send to Server**: The message is sent across the network to the server.
4. **Server Stub:** The server stub unpacks the message and calls the actual server procedure.
5. **Execution & Return:** The server runs the procedure and returns the result to the stub.
6. **Back to Client:** The server stub sends the result back, and the client stub unpacks it.

How to Make a Remote Procedure Call?



The calling environment is suspended, procedure parameters are transferred across the network to the environment where the procedure is to execute, and the procedure is executed there. When the procedure finishes and produces its results, its results are transferred back to the calling environment, where execution resumes as if returning from a regular procedure call.

Note : *RPC is especially well suited for client-server (e.g. query-response) interaction in which the flow of control alternates between the caller and callee. Conceptually, the client and server do not both execute at the same time. Instead, the thread of execution jumps from the caller to the callee and then back again.*

Types of RPC

- **Callback RPC**: Both client and server can act as each other. The server can call back the client, useful in interactive applications. Handles deadlocks and supports peer-to-peer communication.
- **Broadcast RPC**: The client's request is broadcast to all servers. Special broadcast ports are used. Useful when multiple servers can handle the request.
- **Batch-mode RPC**: Groups multiple client requests and sends them together to the server, reducing network overhead. Best for applications with infrequent calls.

What Does RPC do?

- RPC (Remote Procedure Call) lets a program use code on another computer as if it were local.
- A stub acts as a placeholder for the remote code in the client program.
- When the program calls the remote code, the stub sends the request to a local helper.
- The helper finds the remote server and sends the request over the network.
- The server runs the code and sends the result back through its own helpers.
- The client receives the result, making it look like the code ran locally.

Issues of the RPC

1. RPC Runtime: A library that manages the communication in RPC. It handles binding, sending/receiving data, choosing the protocol, and dealing with errors.

2. Stub: A helper code that hides the complexity from the programmer. On the client side, it converts function calls into messages (marshalling/unmarshalling) and works with the runtime to connect to the server. On the server side, the stub provides a similar interface between the run-time system and the local manager procedures that are executed by the server.
3. Binding: The most flexible solution is to use dynamic binding and find the server at run time when the RPC is first made. The first time the client stub is invoked, it contacts a name server to determine the transport address at which the server resides. Binding consists of two parts

- **Naming**: A Server having a service to offer exports an interface for it. Exporting an interface registers it with the system so that clients can use it.
- **Locating**: A Client must import an (exported) interface before communication can begin.

The call semantics associated with RPC

- **Retry Request Message:** Whether to retry sending a request message when a server has failed or the receiver didn't receive the message.
- **Duplicate Filtering:** Remove the duplicate server requests.
- **Retransmission of Results**: To resend lost messages without re-executing the operations at the server side.

Advantages

- **Easy Communication:** RPC lets clients talk to servers using normal procedure calls in high-level programming languages. This makes it simple for programmers to work with.
- **Hidden Complexity:** RPC hides the details of how messages are sent between computers. This means programmers don't need to worry about the underlying network communication.
- **Flexibility:** RPC can be used in both local and distributed environments. This makes it versatile for different types of applications.

Disadvantages

- **Limited Parameter Passing:** RPC can only pass parameters by value. It can't pass pointers, which limits what can be sent between computers.
- **Slower Than Local Calls:** Remote procedure calls take longer than local procedure calls because they involve network communication.
- **Vulnerable to Failures:** RPC depends on network connections, other machines, and separate processes. This makes it more likely to fail than local procedure calls.

RPC vs REST

RPC and REST are two ways to make computer programs talk to each other over the internet. They're different, but both are useful. RPC is good for some things, and REST is good for others. Some companies need RPC, while others prefer REST. Sometimes, developers use both RPC and REST in the same project, but not in the same part of the program. RPC is an old idea, but new versions like gRPC and DRPC are making it popular again. Developers are still using and improving these new types of RPC.
Note: *It's hard to say which one is better - RPC or REST. They're both good when used the right way. The best choice depends on what you're trying to do with your program.*

Data Compression Techniques

The modern digital world is driven by data compression which is a very essential technique where saving and transmitting information efficiency is the most important thing. With the rise in data generation capacity, there have been more difficulties in managing and storing it effectively. This process decreases files sizes such that space for storage is saved while at the same time ensuring higher speeds of transferring the data across networks. The importance of this approach has become even more pronounced with current multimedia applications, cloud storage systems, and mobile communication units that are characterized by limited bandwidth as well as stringent storage conditions.
Data compression can be applied to text, pictures, audio and video among other forms of data. This concept involves minimizing redundancies and irrelevances in representing data so that less bytes or bits are

required to store or convey it. A number of techniques accomplish this purpose via embedding patterns and redundant data within them. Later on we will proceed to a more detailed description about what makes up data compression including different kinds, mechanisms involved as well as its application.

Definition of Data Compression

Data compression is defined as the process whereby information is encoded in less bits than it had originally occupied. This mainly happens through methods that eliminate duplication and other extraneous information.

Compression techniques are useful for reducing file sizes for storage, minimizing bandwidth during transmission and enabling faster uploading/downloading of web content over the internet.

Data Compression Techniques
Data compression can be divided into two categories: lossless and lossy.

Lossless Data Compression

Lossless data compression guarantees that the decompressed data is identical to the original data. It works best for text and data files where precision matters.

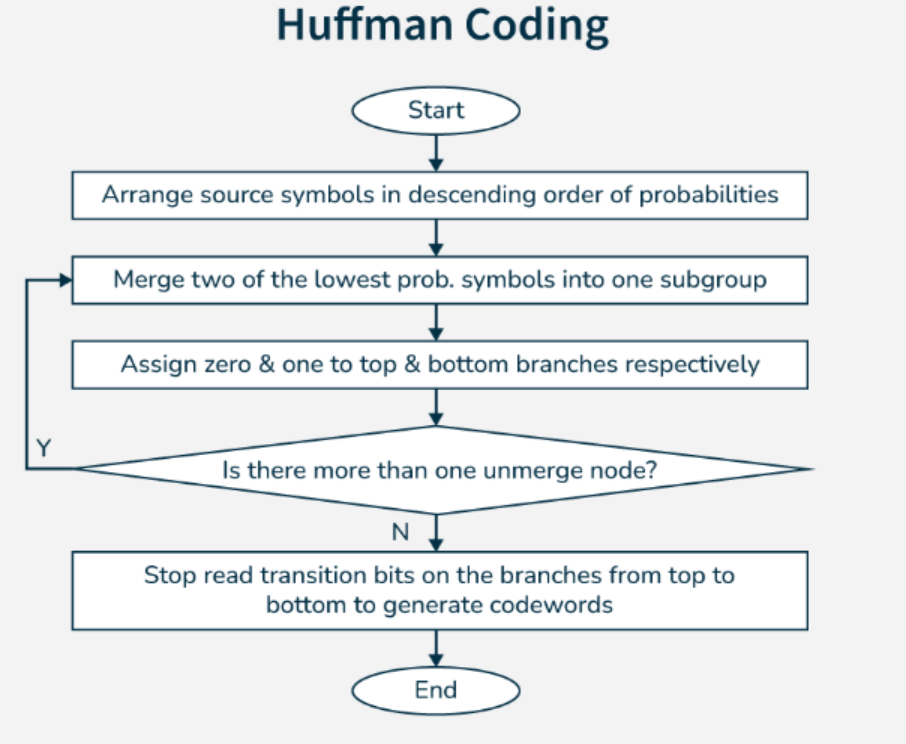
- Huffman coding: Uses a frequency-sorted binary tree to locate values efficiently.
- Run-length encoding (RLE): This compresses sequences of replicated data values.
- Lempel-Ziv-Welch (LZW): It creates a dictionary of data patterns and replaces them with shorter codes.

Lossy Data Compression

Lossy data compression gives away the accuracy of some of its input data for a better compression ratio. It is usually applied to multimedia files, where some loss of detail can be tolerated. Some techniques include:

- Transform Coding: Uses mathematical transforms that shrink the data, usually in JPEG
- Quantization: Reducing the precision of data; it is common in image and video compression.

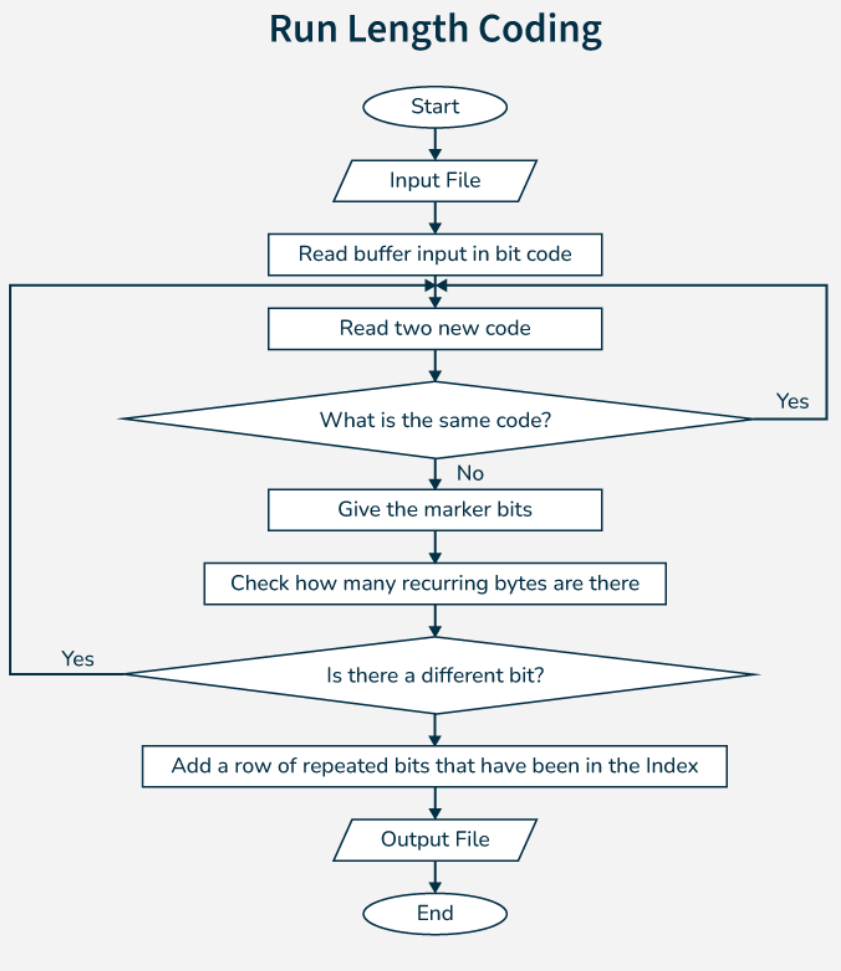
Explanatory diagram for Huffman Coding



Huffman Coding

The algorithm for Huffman coding will create a binary tree in which the more used symbols will have the more minor codes. The diagram shows the tree structure for which we can encode characters according to their frequencies.

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In RLE, repeated data sequences are substituted with a single data value and count. The above diagram depicts how a sequence of repetitive values is compressed to make it small in file size.

Working Principle of Data Compression

There are two main processes that underlie the working principle of data compression:

- Encoding: This is a process in which existing data is examined for patterns, redundancies and irrelevant information. Data is then encoded according to the analysis made, so that it has fewer bits with similar contents.
- Decoding: The compressed data can be restored almost to its original (in lossy) or the original form itself. The result of decompression, in lossless compression, is identical to the original. But, in case of lossy compression though, this data will be decompressed while retaining the most important features only, but with some loss in detail.

Parts of a Compression System

- Encoder: A device used for converting initial information into compressed format.
- Decoder: It restores compressed information back to its initial state.
- Compression Algorithm: Actual compression takes place using this algorithm e.g., Huffman coding and JPEG compression.
- Dictionary: Some algorithms such as LZW have dictionaries where they keep data patterns.

Development of Compression Systems

- Data Input: This refers to original data that needs to be compressed.
- Pattern Recognition: It identifies and analyzes the patterns present or redundancies.
- Encoding: The data is further processed here using the chosen algorithm for compression.
- Storage or Transmission: This point will store the compressed data or pass it through a network.

- Decoding: This is a process where compression is reversed to recover the original or approximately the same data again.

Important Terminologies

- Entropy: This term measures how random or unpredictable data are. Thus higher entropy means more redundancy and therefore lesser entropy favors better compression.
- Compression Ratio: Given by size of compressed data divided by original size of data.
- Bit rate shows how many bits are used per each datum, which affects either good quality with small amount of lossy compressed or large amount of lossy compressed data over lossy compression technologies.

Examples of Data Compression

- Text compression: Where text files are compressed using algorithms, such as Huffman coding so that the size decreases.
- Image Compression: The JPEG format compresses images by reducing resolution and color depth.
- Audio Compression: MP3 format compresses audio by eliminating inaudible frequencies.

Differentiating Features Between Lossy and Lossless Compression

Feature	Lossy Compression	Lossless Compression
Definition	Reduces file size by permanently eliminating some information.	Reduces file size without losing any information.
Data Loss	Some data is lost, potentially reducing quality.	No data is lost; original data can be perfectly reconstructed.
Compression Ratio	Higher compression ratios, leading to smaller file sizes.	Lower compression ratios, resulting in larger file sizes compared to lossy compression.
Quality	Quality can be significantly reduced, especially at high compression levels.	Original quality is preserved with no loss in data.
Common Uses	Used for multimedia data (images, audio, video) where perfect reproduction is not critical.	Used for text, executable files, and other data where exact replication is essential.
Examples	JPEG (images), MP3 (audio), MP4 (video)	PNG (images), FLAC (audio), ZIP (general files)

Advantages of Data Compression

- Reduced Storage: Compressing files saves significant storage space on devices and servers.
- Faster Transmission: Smaller files transfer more quickly, enhancing download and upload speeds.
- Cost Efficiency: Lower costs for storage and bandwidth, beneficial for managing large data volumes.
- Improved File Management: Smaller files are easier to organize, particularly in archives.

- Enhanced Security: Compression often includes encryption, adding an extra layer of data protection.

Disadvantages of Data Compression

- Quality Loss (Lossy): Lossy compression can reduce quality, making it unsuitable for critical applications.
- Decompression Complexity: Requires additional computational resources, slowing down data retrieval.
- Compatibility Issues: Not all platforms support all compression formats, leading to sharing difficulties.
- Data Corruption Risk: Compressed files are more prone to corruption, risking data integrity.
- Additional Processing: Compression and decompression introduce processing overhead, affecting system performance.

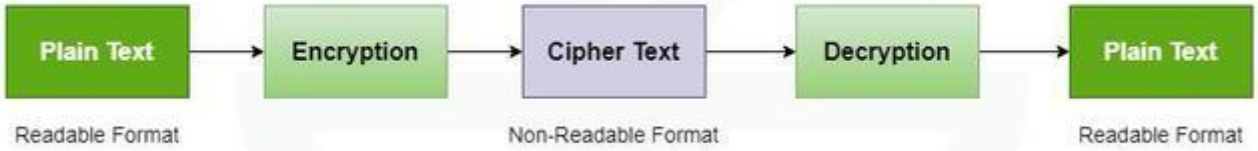
Applications of Data Compression

- Multimedia Files: Compressing audio, video and image files to reduce size while maintaining quality, e.g. MP3 and JPEG formats.
- Web Content Delivery: Reducing the web page sizes to enable faster loading times hence better user experience for instance GZIP compression on HTML and CSS files.
- Email Attachments: This can be done through compressing large attachments to fit within email size limits like ZIP files.
- Remote Sensing: Satellite data compression which enhances its storage and transmission efficiency such as wavelet compression for high-resolution images.
- Medical Imaging: These are medical images that are compressed in order to occupy less space and also ease the transmission process without losing diagnostic details like DICOM format for medical scans.
- Backup and Recovery: Data is compressed during backup so as to save on space thus speeding up recovery since compressed archives can be used for storing data.
- Database Management: The records of databases can be compressed so as to improve query performance as well as save on storage whereby columnar data compression is an example of this.
- Streaming Services: Video or audio streaming services where video or audio is compressed such that minimal bandwidth usage occurs like H.264 codec for video streaming.
- Mobile Applications: Such applications require the reduction of app size to make them download quickly therefore conserve the storage space by compressing images and other information that comes with it.
- Document Management: Text documents together with PDFs are compressed by converting them into a smaller file size that makes it easy to handle, share e.g., using PDF compression tools.

Cryptography and its Types

Cryptography is the science of protecting information using mathematical techniques to ensure confidentiality, integrity, and authentication. It transforms readable data into unreadable form, preventing unauthorized access and tampering.

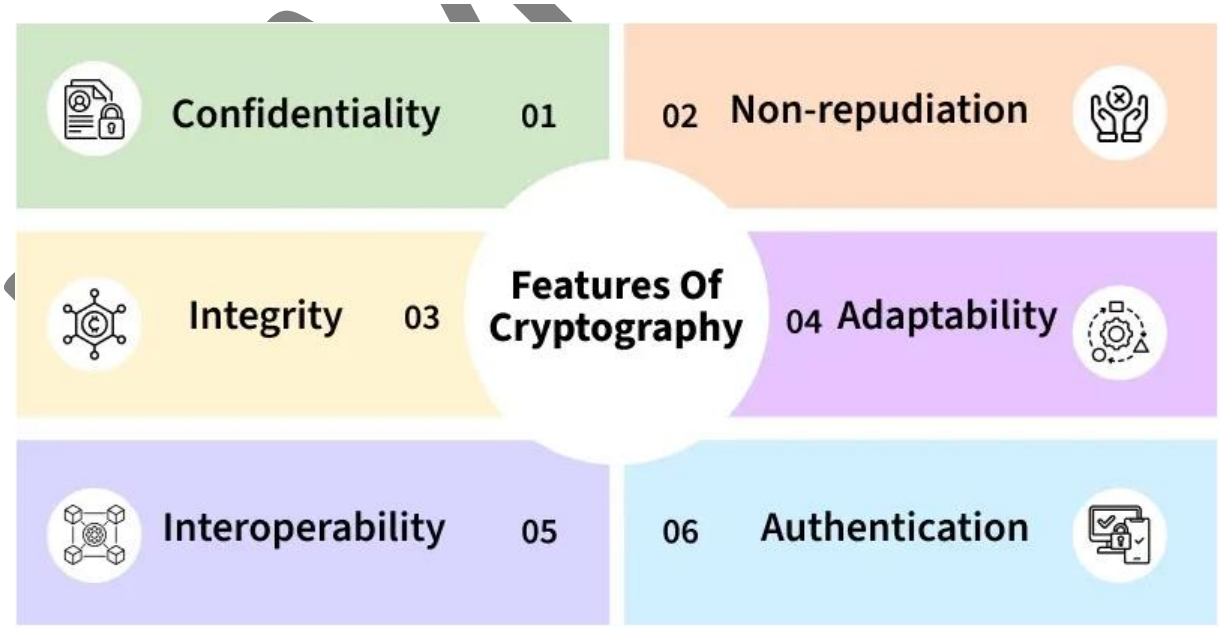
- Converts plaintext into ciphertext using algorithms and keys
- Ensures confidentiality, integrity, authentication, and non-repudiation
- Used in secure communication, digital signatures, passwords, and online transactions



These algorithms are used for cryptographic key generation, digital signing, and verification to protect data privacy, web browsing on the internet and to protect confidential transactions such as credit card and debit card transactions.

Features Of Cryptography

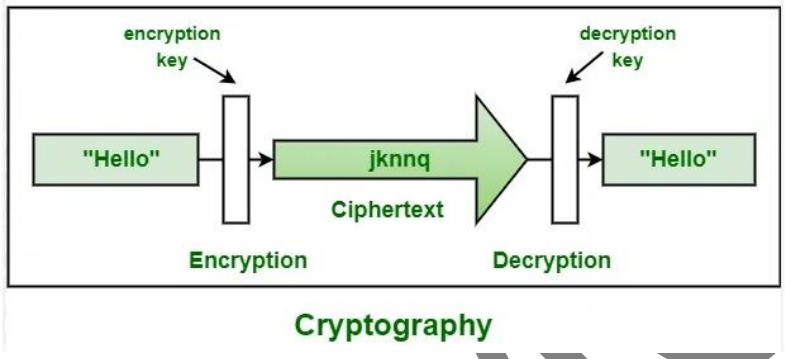
The features of cryptography that makes it a popular choice in various applications could be listed down as



1. **Confidentiality:** Information can only be accessed by the person for whom it is intended and no other person except him can access it.
2. **Non-repudiation:** The creator/sender of information cannot deny his intention to send information at a later stage.
3. **Integrity:** Information cannot be modified in storage or transition between sender and intended receiver without any addition to information being detected.

4. **Adaptability:** Cryptography continuously evolves to stay ahead of security threats and technological advancements.
5. **Interoperability:** Cryptography allows for secure communication between different systems and platforms.
6. **Authentication:** The identities of the sender and receiver are confirmed. As well destination/origin of the information is confirmed.

Working of Cryptography
As we all know that cryptography technique is use to convert plain text into ciphertext. This technique is done by cryptographic key. Basically cryptographic key is a string of characters which is used to encrypts the data and decrypt the data.



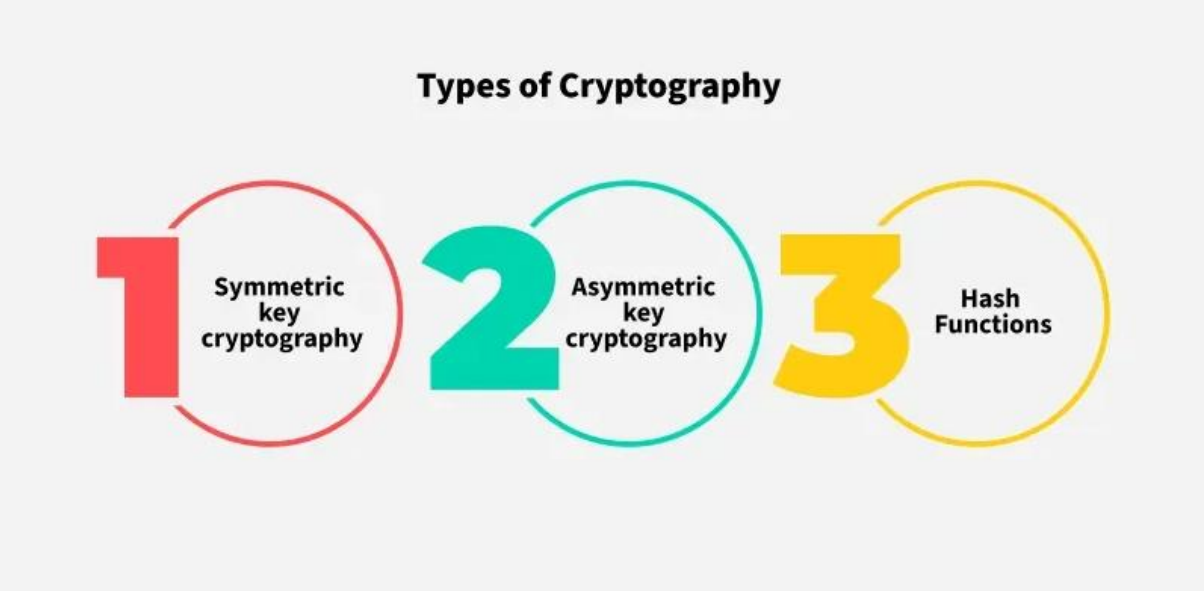
Here,

"Hello" is a plaintext and convert into ciphertext "jknnq" with the help of cryptographic key and then decrypt into "Hello".

- **Plaintext is created**
  The original readable message or data.
- **A cryptographic algorithm is chosen**
  Example: AES, RSA, SHA-256, etc.
- **A key is applied**
  This key controls how the encryption or hashing is performed.
- **Encryption converts plaintext → ciphertext**
  The data becomes unreadable to unauthorized users.
- **Ciphertext is transmitted or stored securely**
  Even if intercepted, it cannot be understood without the correct key.
- **Decryption converts ciphertext → plaintext**
  The receiver uses the correct key (same key for symmetric, different for asymmetric).
- **Integrity checks may be used**
  Hashing ensures the message has not been altered.

Types Of Cryptography

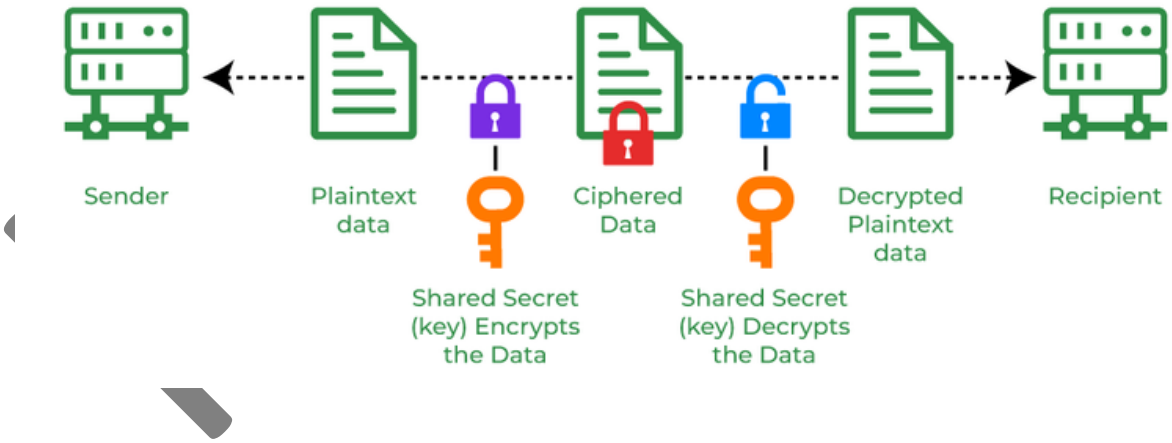
There are three types of cryptography, namely Symmetric Key Cryptography, Asymmetric Key Cryptography and Hash functions, here's a detailed explanation below:



1. Symmetric Key Cryptography

Symmetric Key Cryptography is an encryption system where the sender and receiver of a message use a single common key to encrypt and decrypt messages.

- Symmetric Key cryptography is faster and simpler but the problem is that the sender and receiver have to somehow exchange keys securely.
- The most popular symmetric key cryptography systems are Data Encryption Systems (DES) and Advanced Encryption Systems (AES) .



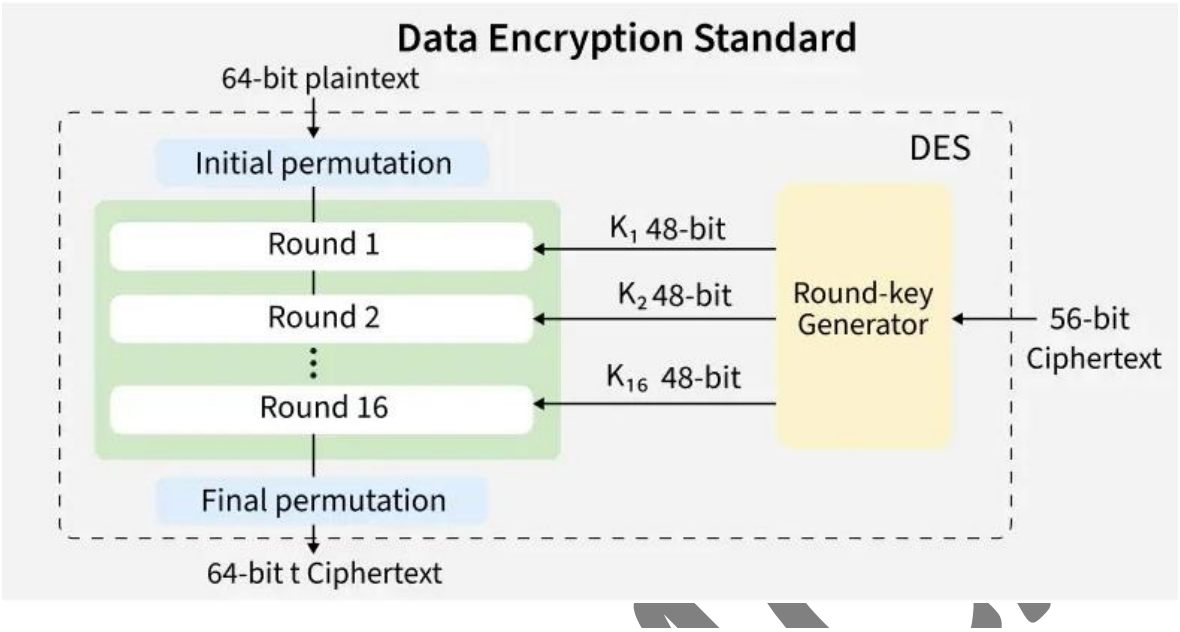
Example:

Data Encryption Standard (DES)

DES (Data encryption standard) is an older encryption algorithm that is used to convert 64-bit plaintext data into 48-bit encrypted ciphertext.

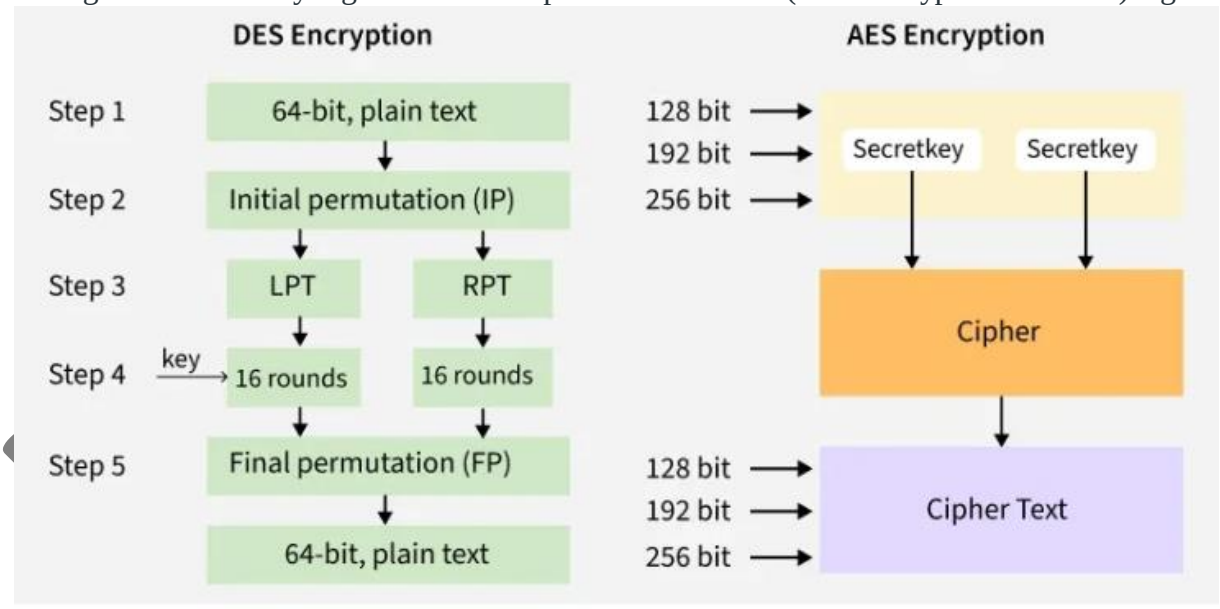
- It uses symmetric keys (which means same key for encryption and decryption).

- It is kind of old by today's standard but can be used as a basic building block for learning newer encryption algorithms.



AES (Advanced Encryption Standard):
AES s a popular encryption algorithm which uses the same key for encryption and decryption.

- It is a symmetric block cipher algorithm with block size of 128 bits, 192 bits or 256 bits.
- AES algorithm is widely regarded as the replacement of DES (Data encryption standard) algorithm.



Example for Encryption and Decryption of AES tool.

2. Hash Functions
Hash functions do not require a key. Instead, they use mathematical algorithms to convert messages of any arbitrary length into a fixed-length output, known as a hash value or digest.

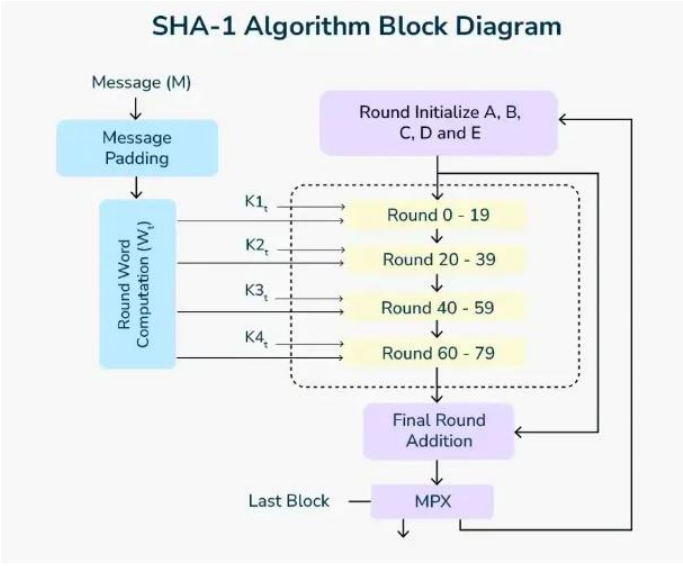
Hash functions are designed to be one-way, meaning the original input cannot be derived from the output.

Given Below, Some of the most widely used hash functions include:

SHA-1:

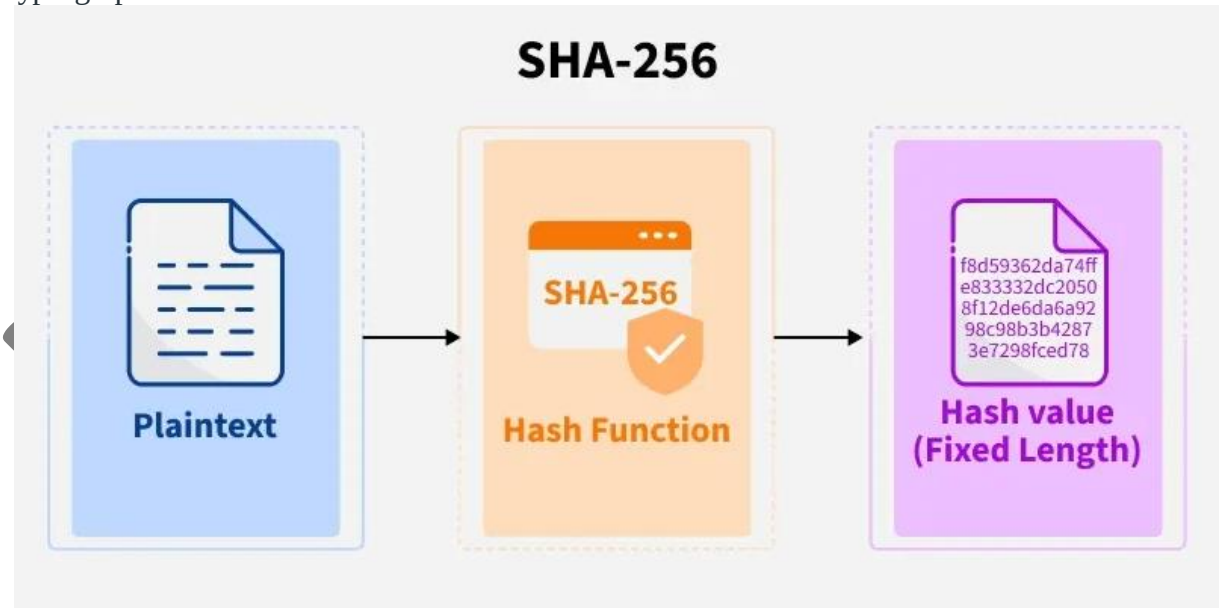
- SHA-1 stands for Secure Hash Algorithm 1.

- It is a cryptographic hash function developed by the National Security Agency (NSA) and published by NIST in 1995.
- It was designed to ensure data integrity by converting any input into a fixed-size 160-bit (20-byte) hash value.



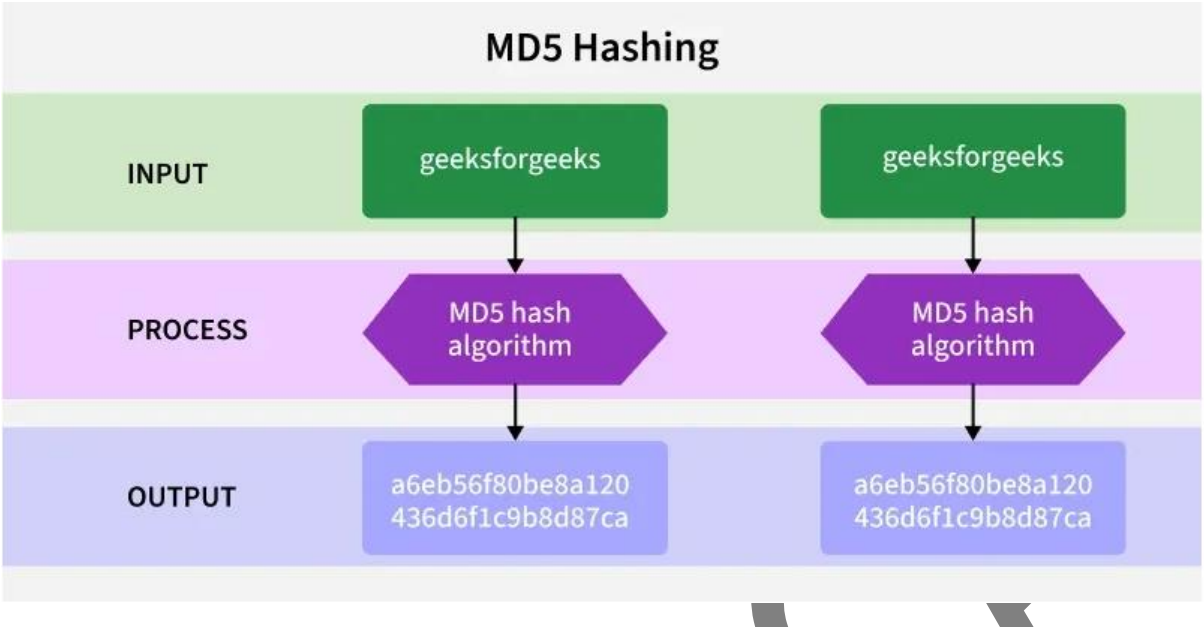
SHA-256:

- The SHA-256 algorithm belongs to the family of the SHA 2 algorithms.
- It generates a fixed 256-bit (32-byte) hash value from input data of any length.
- It is widely used in various security applications, including data integrity verification, digital signatures, blockchain technology, and password hashing, due to its strength and resistance to known cryptographic attacks.



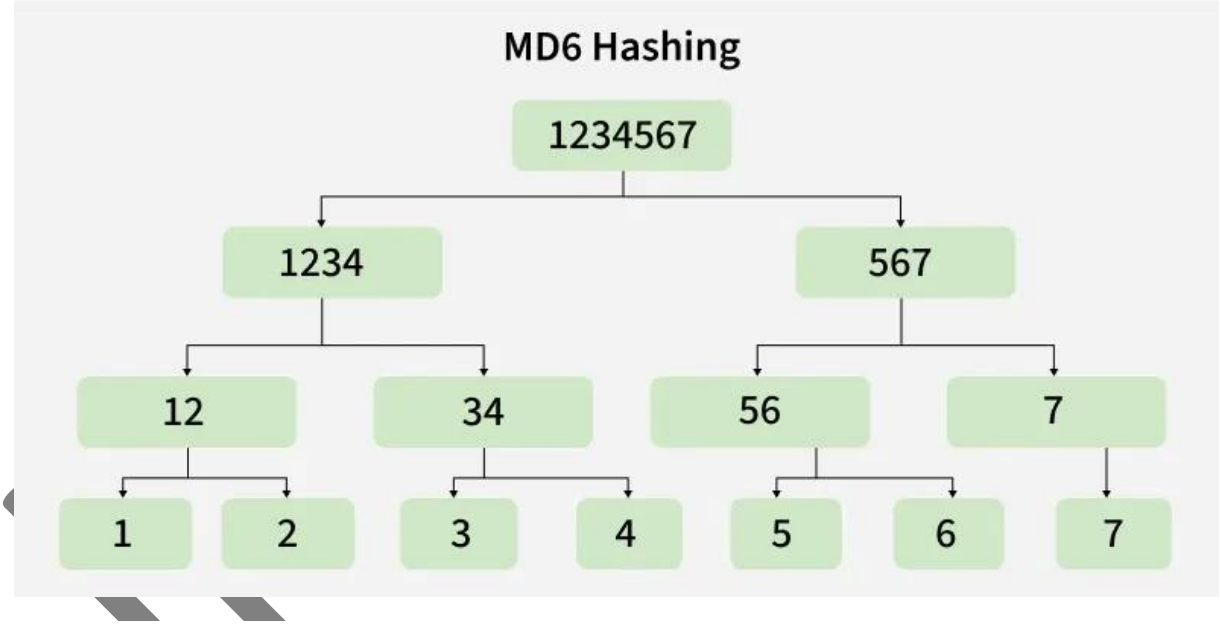
MD5:

- MD5 produces a 128-bit hash value (32-character hexadecimal number) from input data of any size.
- Originally designed for data integrity verification, MD5 was widely used for checksums, digital signatures, and file verification.



MD6:

- MD6 is a cryptographic hash function designed by Ronald Rivest and his team in 2008 as a successor to the MD5 and MD4 algorithms.
- It was created to be highly secure and suitable for modern computing systems, including multi-core processors.



3. Asymmetric Key Cryptography

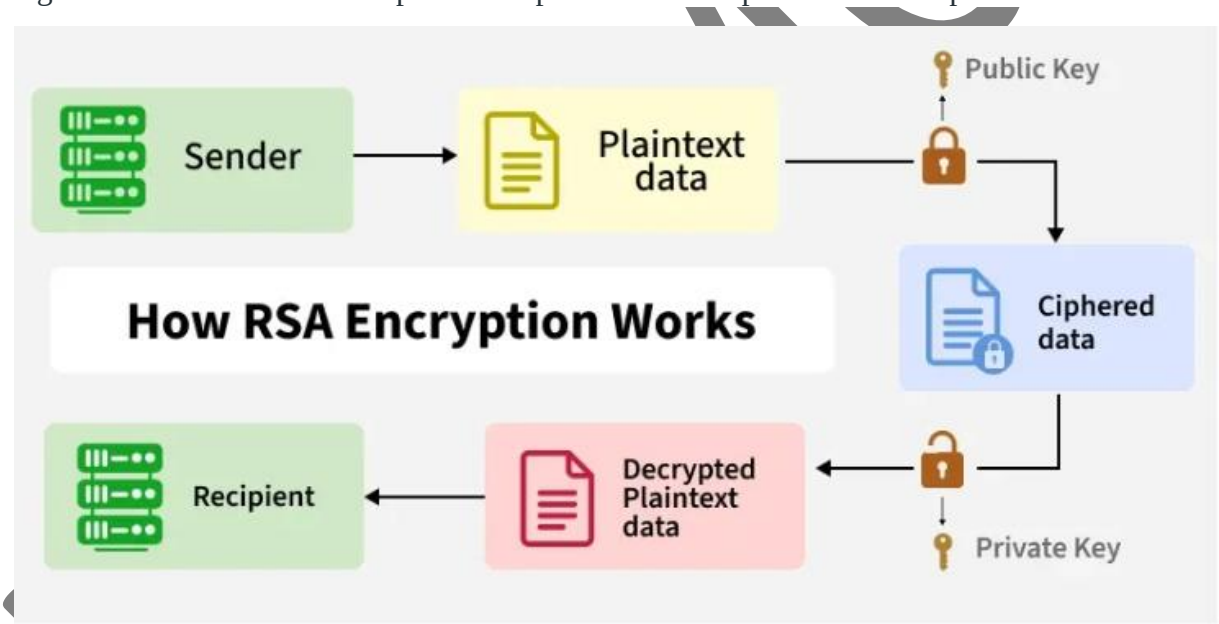
In Asymmetric Key Cryptography a pair of keys is used to encrypt and decrypt information. A sender's public key is used for encryption and a receiver's private key is used for decryption. Public keys and Private keys are different. Even if the public key is known by everyone the intended receiver can only decode it because he holds his private key. The most popular asymmetric key cryptography algorithm is the RSA algorithm.



Examples:

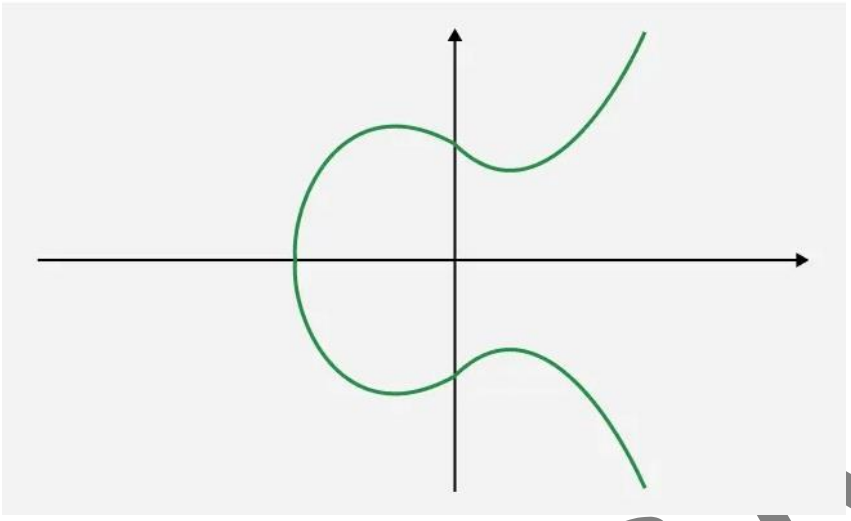
RSA (Rivest-Shamir-Adleman):

RSA is an basic asymmetric cryptographic algorithm which uses two different keys for encryption. The RSA algorithm works on a block cipher concept that converts plain text into cipher text and vice versa.



Encryption in RSA

ECC (Elliptic Curve Cryptography):
lliptic Curve Cryptography (ECC) is a type of asymmetric encryption that provides strong security with smaller keys than RSA. It's efficient, fast, and ideal for devices with limited resources like smartphones, IoT devices, and blockchain wallets. ECC is widely used in secure communications such as TLS/SSL and cryptocurrencies due to its lightweight yet powerful encryption.



Applications of Cryptography

Cryptography has wide area of applications in the modern world, where the technology is rapidly evolving. These are some of the most common applications of cryptography listed below:



1. Password Security: Passwords are hashed before storage to prevent exposure even if the database is leaked.
2. Digital Currencies: Bitcoin and other cryptocurrencies use cryptography for transaction integrity and identity protection.
3. Secure Web Browsing (HTTPS): SSL/TLS protocols encrypt communication between browser and server.
4. Electronic Signatures: Legally valid digital signatures created using public-key cryptography.
5. Authentication Systems: Used in bank logins, secure networks, and access control systems.
6. End-to-End Encryption: Messaging apps like WhatsApp and Signal ensure only intended users read the messages.

File Transfer Protocol (FTP)

File transfer protocol (FTP) is an Internet tool provided by TCP/IP. It helps to transfer files from one computer to another by providing access to directories or folders on remote computers and allows software, data and text files to be transferred between different kinds of computers.

- It encourages the direct use of remote computers.
- It shields users from system variations (operating system, directory structures, file structures, etc.)
- It promotes the sharing of files and other types of data.

Note: *The end-user in the connection is known as localhost and the server which provides data is known as the remote host.*

Applications of FTP

- **Business File Sharing**: Used by large enterprises to share files between employees across locations.
- **Backup & Recovery**: IT companies use FTP to maintain disaster recovery backup sites.
- **Financial Sector**: For secure transmission of sensitive documents between institutions and regulatory authorities.
- **Collaboration**: Employees use FTP to share files and project data with co-workers.

FTP Client and Server Model
FTP follows a client-server architecture.

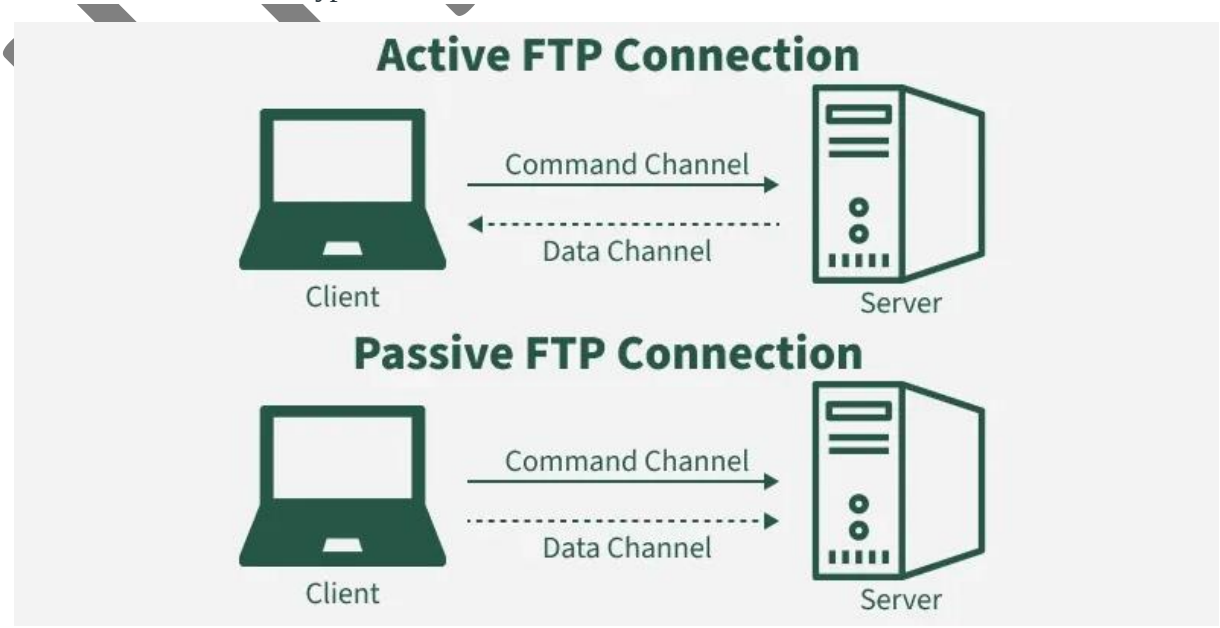
- **FTP Client**: A program that runs on the user's computer, enabling communication with an FTP server. It provides commands to connect, browse and transfer files.
- **FTP Server**: A system that hosts the files and directories, waiting for requests from clients.

Commands

- get the filename(retrieve the file from the directories get server)
- mget filename(retrieve multiple files from the server)
- ls(lists files available in the current directory of the server)

Note: *Many modern operating systems have built-in FTP programs, making file transfer easier without needing to memorize commands.*

Types of FTP Connections
FTP connections are of two types:



1. Active FTP connection

- In an Active FTP connection, the client establishes the command channel and the server establishes the data channel.
- When the client requests the data over the connection the server initiates the transfer of the data to the client.
- It is not the default connection because it may cause problems if there is a firewall in between the client and the server.

2. Passive FTP connection

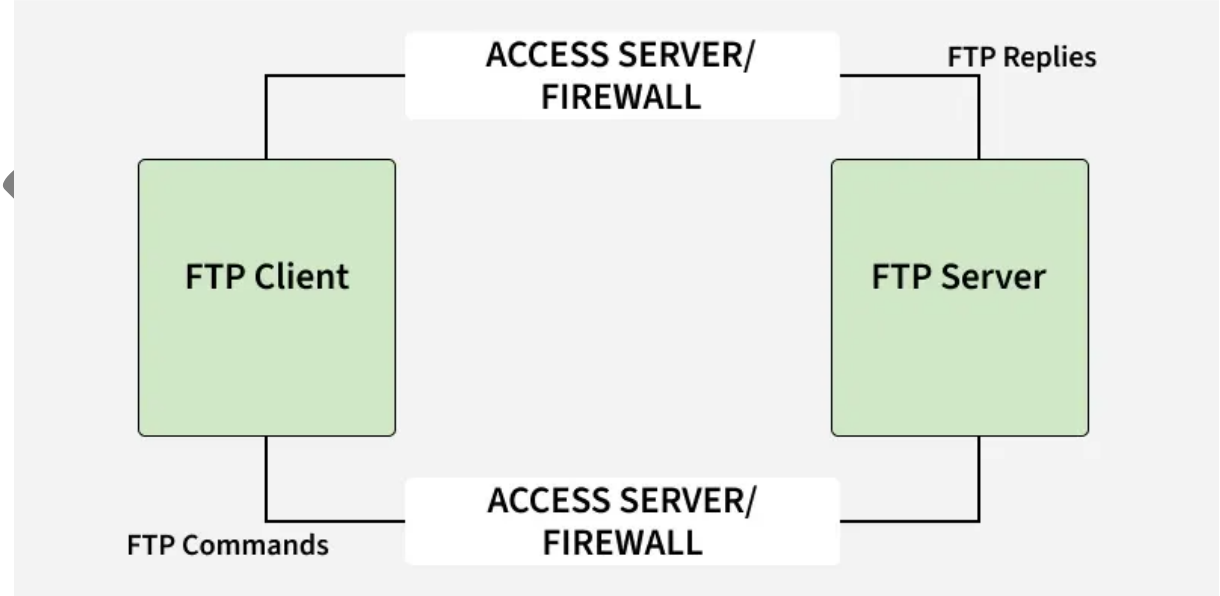
- n a Passive FTP connection, the client establishes both the data channel as well as the command channel.
- When the client requests the data over the connection, the server sends a random port number to the client, as soon as the client receives this port number it establishes the data channel.
- It is the default connection, as it works better even if the client is protected by the firewall.

Anonymous FTP
Some servers provide anonymous FTP, where files are available for public access without authentication.

- The username is set to anonymous.
- The password is typically the user's email address.
- Access is limited: users can download files but not browse directories or make modifications.

How FTP works?
When an FTP connection is established, two parallel channels are created:

1. **Command Channel**: Used for transmitting commands (like login, navigation and file requests). Operates over Port 21.
2. **Data Channel**: Used for transferring the actual data/files. Operates over Port 20.



Detail Steps of FTP

- Client contacts the server on Port 21.
- Authentication is performed (username/password or anonymous login).
- The client browses directories using commands.

- When a file transfer is requested, the server opens a new data connection to the client.
- After transferring a file, the data connection closes (but the control connection remains open).
- The process repeats for additional file transfers.

***Note**: FTP uses the NVT ASCII character set for communication over the control channel, similar to TELNET and SMTP.*

Transmission Mode
FTP supports three transmission modes:

1. **Stream Mode**: (Default) Data is sent as a continuous stream of bytes. TCP handles fragmentation. Connection closes automatically at the end of transmission.
2. **Block Mode**: Data is sent in blocks, each with a 3-byte header describing the block. Useful for structured data.
3. **Compressed Mode**: Large files are compressed before transfer to save bandwidth and speed up transmission.

FTP Commands

Command	Meaning
cd	Change working directory on the remote host.
close	Close the FTP connection.
quit	Exit the FTP session.
pwd	Show current working directory on remote host.
dir/ls	List files in the current directory.
help	Display list of client FTP commands.
remotehelp	Display list of server FTP commands.
type	Specify file type (ASCII/Binary).
struct	Specify file structure.

Why FTP?
While other protocols like HTTP can also transfer files, FTP provides a more specialized, reliable and standardized way to handle file exchanges between heterogeneous systems.

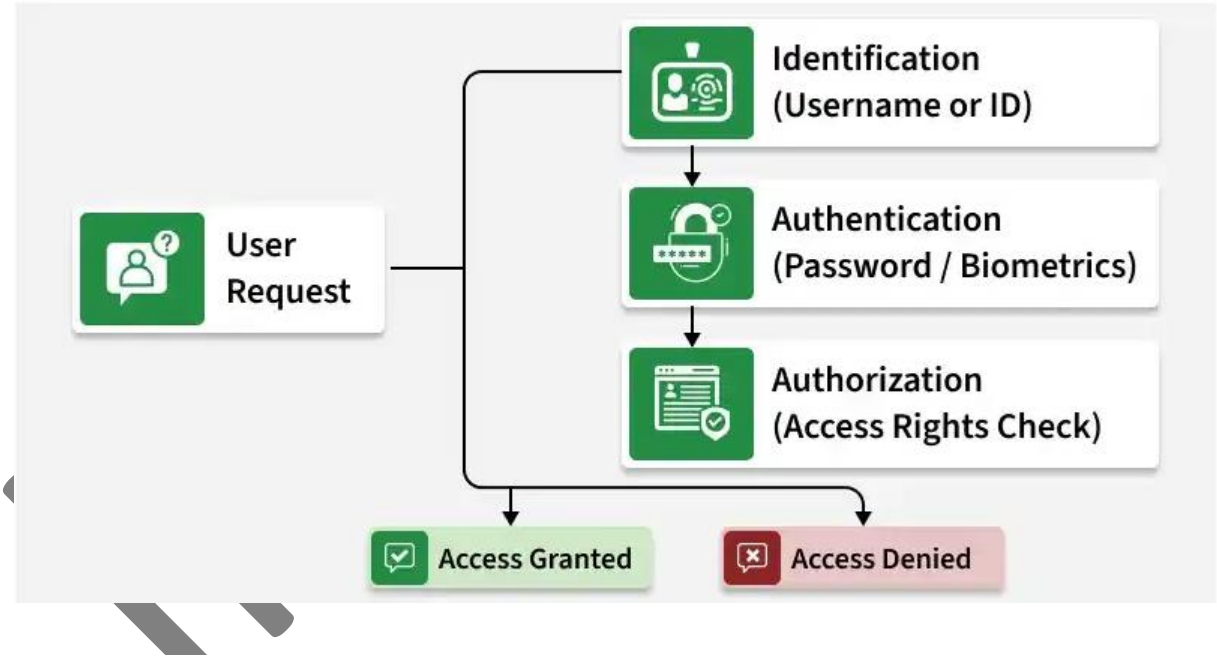
FTP supports transferring:

- ASCII files (default, character-based).
- EBCDIC files (for mainframes).
- Binary (image) files (default for non-text files like executables, images, etc.).

Access Control in Computer Network

Access Control is the process of restricting access to systems, networks, or resources based on predefined security policies. It determines who can access specific resources and what actions they can perform. Access control systems verify user identity using credentials such as passwords, PINs, biometric scans, or security tokens, helping prevent unauthorized access. Its key features include:

- **Identity Verification:** Ensures that the user attempting to access the system is legitimate.
- **Permission Management:** Defines what actions an authorized user is allowed to perform within the system.
- **Security Enforcement:** Prevents unauthorized users from accessing sensitive resources or data.
- **Policy-Based Access:** Grants or restricts access according to predefined organizational security policies.



Authentication Factors

- Password or PIN
- Bio-metric measurement (fingerprint & retina scan)
- Card or Key

For computer security, access control includes the authorization, authentication, and audit of the entity trying to gain access. Access control models have a subject and an object.

Components of Access Control

- **Authentication:** Authentication is the process of verifying the identity of a user. User authentication is the process of verifying the identity of a user when that user logs in to a computer system.
- **Authorization:** Authorization determines the extent of access to the network and what type of services and resources are accessible by the authenticated user. Authorization is the method of enforcing policies.
- **Access:** After the successful authentication and authorization, their identity becomes verified, This allows them to access the resource to which they are attempting to log in.
- **Manage:** Organizations can manage their access control systems by adding or removing user accounts, authentication methods, and access permissions as required. Managing these systems can be difficult in modern IT setups that combine cloud services and physical systems.
- **Audit:** The access control audit method enables organizations to follow the principle. This allows them to collect data about user activities and analyze it to identify possible access violations.

Working of Access Control

Access control works by verifying a user's identity and granting permissions based on predefined security policies. Users provide credentials such as passwords, PINs, security tokens, or biometric data to authenticate their identity. In many systems, Multi-Factor Authentication (MFA) is used to increase security by requiring more than one verification method. Once the user's identity is confirmed, the system applies access control policies to determine what resources the user can access and what actions they are allowed to perform.
The process generally includes the following steps:

1. **Identification:** User provides credentials such as a username or ID.
2. **Authentication:** System verifies the user's identity using passwords, biometrics, or tokens.
3. **Authorization:** System determines the level of access granted to the user.
4. **Access Granting:** User is allowed to access permitted resources based on security policies.

Types of Access Control

- **Attribute-based Access Control (ABAC):** In this model, access is granted or declined by evaluating a set of rules, policies, and relationships using the attributes of users, systems and environmental conditions.
- **Discretionary Access Control (DAC):** In DAC, the owner of data determines who can access specific resources.
- **History-Based Access Control (HBAC):** Access is granted or declined by evaluating the history of activities of the inquiring party that includes behavior, the time between requests and content of requests.
- **Identity-Based Access Control (IBAC):** By using this model network administrators can more effectively manage activity and access based on individual requirements.
- **Mandatory Access Control (MAC):** A control model in which access rights are regulated by a central authority based on multiple levels of security. Security Enhanced Linux is implemented using MAC on the Linux operating system.
- **Organization-Based Access control (OrBAC):** This model allows the policy designer to define a security policy independently of the implementation.
- **Role-Based Access Control (RBAC):** RBAC allows access based on the job title. RBAC eliminates discretion on a large scale when providing access to objects. For example, there should not be permissions for human resources specialist to create network accounts.

- **Rule-Based Access Control (RAC):** RAC method is largely context based. Example of this would be only allowing students to use the labs during a certain time of day.

Different access control models are used depending on the compliance requirements and the security levels of information technology that is to be protected. Access control is of 2 types:

- **Physical Access Control:** Physical access control restricts entry to campuses, buildings, rooms and physical IT assets.
- **Logical Access Control:** Logical access control limits connections to computer networks, system files and data.

Challenges of Access Control

- **Distributed IT Systems:** Current IT systems frequently combine internet and on-premise networks. These systems may be distributed geographically and comprise various devices, assets, and virtual machines. Access is allowed to all of these devices, and keeping track of them can be challenging.
- **Policy Management:** Policy makers within the organization create policies, and the IT department converts the planned policies into code for implementation. Coordination between these two groups is essential for keeping the access control system up to date and functioning properly.
- **Monitoring and Reporting:** Organizations must constantly check access control systems to guarantee compliance with corporate policies and regulatory laws. Any violations or changes must be recognized and reported immediately.
- **Access Control Models:** Access control mechanisms provide varying levels of precision. Choosing the right access control strategy for your organization allows you to balance acceptable security with employee efficiency.

Types of Authentication Mechanism

- Two-factor authentication
- Multi-factor authentication
- One-time password
- Three-factor authentication
- Biometrics
- Hard Tokens
- Soft Tokens
- Contextual Authentication
- Device identification

Methods for Implementing Access Control

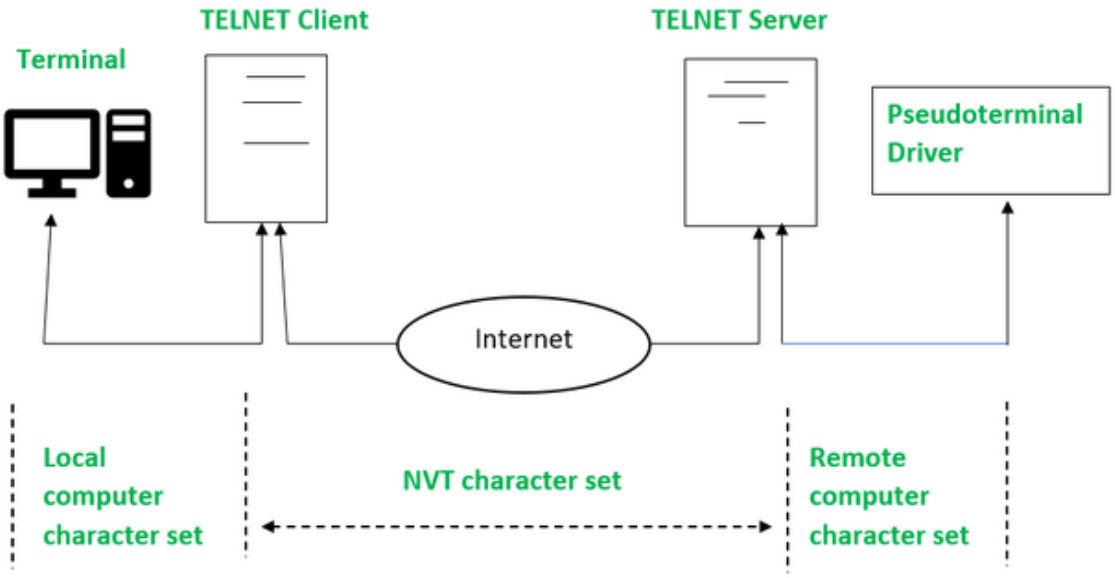
- **Virtual Private Networks (VPNs):** VPNs allow users to securely access organizational networks remotely by encrypting internet connections. However, excessive use may sometimes cause performance issues such as latency.
- **Other Methods:** Additional techniques include identity repositories, monitoring and reporting tools, password management systems, provisioning tools, and security policy enforcement services that help manage and enforce access permissions effectively.

What is Network Virtual Terminal in TELNET?

TELNET stands for Terminal Network. It follows the approach of client-server. The connection of the local system to the remote system is established in such a way that the local system starts to appear in the remote system. This TCP/IP protocol is used to provide virtual terminal service. Therefore, it allows users to access the services of the remote system and then get back the results to the local system.

Network Virtual Terminal (NVT):

- NVT (Network Virtual Terminal) is a virtual terminal in TELNET that has a fundamental structure that is shared by many different types of real terminals.
- TELNET clients and servers send data, commands, and server output using a virtual terminal type called NVT, rather than a variety of terminal types for different operating systems (Network Virtual Terminal) as systems are heterogeneous nowadays.
- The character set, carriage return, and line termination are all defined by the NVT (Network Virtual Terminal).
- NVT (Network Virtual Terminal) was created to make communication viable between different types of terminals with different operating systems.



How does NVT work?

It works as follows and is also represented in the above figure:

- The client TELNET converts the characters from the local terminal into NVT format and then sends them to the network.
- Then the server TELNET converts NVT-formatted data and commands into a format that the Remote Computers can understand.

Network Virtual Terminal (NVT) Character Set:

- The Network Virtual Terminal (NVT) primarily employs two sets of characters: one for data and another for control.
- The NVT is an 8-bit character set for data, with the 7 lowest-order bits identical to ASCII and the highest bit set to 0.
- The NVT uses an 8-bit bit character set to communicate control characters between the computers, with the highest-order bit set to 1.
- For sending data and control characters the TELNET makes use of the same connection by just inserting control characters into the data stream.
- Each control character is preceded by the Special Control character, which is popularly known as Interpret as Control, for separating the data characters from the control characters (IAC).

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Question Bank Unit II: -

1. How RPC Executes in Session Layer? Explain with suitable example. (8 marks)
2. What do you mean by Connection Management? (8 marks)
3. What are the design Issues in application layer? (8 marks)
4. Explain design issues in Presentation Layer? (8 marks)
5. Write a note on Virtual terminal. (4 marks)
6. What is Cryptography? Explain the techniques and types with three examples. (8 marks)
7. What are the design issues of transport layer? Explain working of Transport layer. (8 marks)
8. Write any two elements of Transport Protocol. (8 marks)
9. What are the different Data Compression Techniques? Explain. (8 marks)
10. Explain the Connection Management of TCP. (4 marks)

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